

Ad Media, Cohort Targeting & Self-Learning Targeting — Design & Compliance Note

Prepared for the owner and counsel. Describes the advertiser ad capabilities added this session: **audio/video creatives** (interstitial + PPC network), **cohort targeting** from the mandatory Know-Your-Customer (KYC) survey across every ad surface, a **self-learning AI targeting layer** matching the platform's AI posture, and the **gap fixes** that make targeting consistent everywhere. **Not legal advice** — the privacy/advertising questions flagged below are for counsel. Current as of 2026-09-11.

1. What was added (summary)

1. **Audio & video ad creatives.** Advertisers may supply an **image, video, or audio** creative — for the full-screen **interstitial** placements and the **PPC ad grid**. **The PPC grid is now a full-screen experience:** the shopper taps **one** tile and it opens a full-screen takeover where the advertiser's video/audio **loops through a 30-second watch gate** (30s × 16 ads = 8 minutes total); the questions then appear beneath the still-looping ad, the shopper submits (credited server-side), the real product Buy-Now page reveals, and they **swipe (or use on-screen arrows) straight to the next ad** — they tap the grid only once.
 2. **Cohort targeting from the KYC survey — on every ad surface.** Advertisers can target a specific cohort from the first-party welcome-survey answers. Targeting now applies to **(a) the in-app interstitial ads, (b) the PPC ad grid, (c) the social-media (endorser) distribution, and (d) the premium PPC auto-advertise engine.**
 3. **A self-learning AI targeting layer.** An AI layer learns which creatives resonate with which cohorts and **orders matching ads by learned relevance** for each user, and **recommends** cohort refinements — under the platform's graduated-autonomy posture with a permanent gate and a kill switch.
 4. **Gap fixes.** Targeting/personalization now flows through placements that previously missed it (the premium ad-free slot, the PPC grid, the auto-advertise engine).
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2. Audio/video creatives

- A creative carries `media_type` (`image` | `video` | `audio`), `media_url`, and `poster_url` (still image; falls back to the thumbnail). Resolved **at serve-time**, so the admin enable-flags (`AD_MEDIA_VIDEO_ENABLED`, `AD_MEDIA_AUDIO_ENABLED`) apply retroactively — turning a format off makes existing creatives fall back to the image, nothing breaks.
- **Interstitial placements** (between-survey + in-app full-screen): the creative plays inside the ad slot with the mandatory **countdown unchanged**; a clip that ends early can unlock "Continue" early (`AD_MEDIA_UNLOCK_ON_END`). Video autoplays **muted with a tap-for-sound** control.
- **PPC ad grid (full-screen takeover):** the shopper taps **one** tile in the grid and it opens a **full-screen takeover**. The advertiser's **video/audio loops through a 30-second watch gate** (30s × 16 ads = 8 minutes total); when the gate clears the **questions appear beneath the still-looping ad**, the shopper submits (the answer is **credited server-side**), the **real product Buy-Now page reveals**, and the shopper **swipes — or uses on-screen arrows — straight to the next ad**. They tap the grid only once. Audio ads loop over the thumbnail/poster the same way. `AdGridSurvey` is the single canonical grid and all entry points lead to it.
- **Rights & disclosure unchanged:** the advertiser attests content rights (DMCA); house branding still applies; the AI-generated disclosure label + C2PA provenance apply only to AI-generated creatives.

3. Cohort targeting — now on every surface

- **Source:** the mandatory KYC welcome survey (`kyc.ts`); answers live on `User.kyc_answers`. Targetable fields: product categories, goals, game genres, monthly budget, shopping style, shopping frequency, device.
- **Advertiser control:** pick any combination of cohort values and a **match mode** — **ANY** (broad) or **ALL** (narrow). No selection = **untargeted → everyone**.
- **Applies to:**
 - **In-app interstitial ads** — the selector filters candidates to matching creatives (house ad fills if none match).
 - **PPC ad grid** (the single canonical `AdGridSurvey` full-screen experience) — the grid feed only shows a targeted PPC ad to users whose answers match.
 - **Social-media (endorser) distribution** — an advertiser's post is queued only to consenting, opted-in members whose answers match (still `#ad`-disclosed, still opt-in).
 - **Premium PPC auto-advertise** — the AI engine posts each advertiser's ad only to consenting members matching that advertiser's cohort.
- One matcher (`ad-targeting.ts`) governs all four, so behavior is identical everywhere.

4. Self-learning AI targeting layer

- **Posture (matches the platform's other AI):** graduated autonomy `suggest` → `assist` → `auto` (`AD_TARGETING_AI_AUTONOMY` , default `assist`), a **permanent gate**, and a **kill switch** (`AD_TARGETING_AI_KILL`). When off/killed, everything falls back to plain cohort matching.
- **What it does:** learns from observed engagement (a shopper answered a PPC creative and marked “interested”, an interstitial completed) which creatives resonate with which cohorts, using **sample-smoothed** rates so a one-off fluke can't outrank a proven cohort. It then **orders matching ads by learned relevance** for the current user (a bias only — it never excludes an ad) and **recommends** cohort refinements to advertisers.
- **The permanent gate (important for counsel):** the AI may reorder relevance freely, but it **never changes an advertiser's contracted targeting, never targets an individual, and never introduces a new or sensitive attribute** — it only reweights the same non-sensitive, first-party KYC cohorts. Applying a recommended targeting change to an advertiser's campaign stays a human/assist action, never automatic.
- **Mechanics:** `adTargetingLearn` (admin/scheduled) rebuilds the model into an `AdTargetingModel` singleton; `adTargetingAiStatus` is an admin read of config + the learned model summary; the selectors consult the model to rank matching creatives.

5. Privacy & compliance posture (for counsel)

- **First-party, self-reported, non-identifying.** All targeting uses only cohort attributes the user provided in the survey. It targets a **group**, never an individual.
- **No sensitive/protected categories.** Current fields are commercial preferences (interests, budget band, style, frequency, device, game genres) — not protected classes. The survey is admin/AI-editable; **counsel should confirm no sensitive-category questions are added** without separate review.
- **Social distribution keeps its guardrails:** members remain **opt-in**, each post carries the **FTC #ad disclosure**, and reach counts only after the member confirms they posted.
- **The AI layer changes relevance ordering, not who is eligible or what an advertiser contracted** — so it does not expand data use beyond the cohort matching already disclosed.
- **For counsel to confirm:** (a) the Privacy Policy / consent disclose that **welcome-survey answers are used to select which ads a user sees and which ads a consenting member may post**, and the data-safety/nutrition-label forms match; (b) state **targeted-advertising opt-out** obligations (CPRA, CO/CT/VA, etc.) are met, with the existing `tracking_opt_out` honored; (c) using survey answers for advertiser targeting stays within the survey's stated personalization purpose; (d) no protected-class or sensitive targeting is offered; (e) the AI ranking layer's data use is covered by the same disclosures.

6. Admin settings (defaults)

Setting	Default	Purpose
<code>AD_MEDIA_VIDEO_ENABLED</code>	ON	Allow advertiser video creatives (else fall back to image)
<code>AD_MEDIA_AUDIO_ENABLED</code>	ON	Allow advertiser audio creatives (else fall back to image)
<code>AD_MEDIA_AUTOPLAY</code>	ON	Autoplay media (video muted + tap-to-unmute)
<code>AD_MEDIA_UNLOCK_ON_END</code>	ON	Unlock “Continue” when an interstitial clip ends early
<code>AD_MEDIA_MAX_SECONDS</code>	60	Advertiser-facing clip-length guidance
<code>AD_TARGETING_ENABLED</code>	ON	Cohort targeting across all ad surfaces
<code>AD_TARGETING_AI_ENABLED</code>	ON	The self-learning relevance/recommendation layer
<code>AD_TARGETING_AI_AUTONOMY</code>	assist	AI autonomy cap: suggest / assist / auto
<code>AD_TARGETING_AI_KILL</code>	OFF	Emergency kill switch for the AI layer
<code>AD_TARGETING_AI_MIN_SAMPLE</code>	30	Min observations before the AI recommends a cohort
<code>AD_TARGETING_AI_PRIOR / _PRIOR_WEIGHT</code>	0.1 / 20	Smoothing prior + weight
<code>AD_TARGETING_LEARN_SAMPLE</code>	5000	Rows the learn pass samples per run

7. Files touched (for the record)

- **New:** `backend/sdk/ad-media.ts` , `backend/sdk/ad-targeting.ts` , `backend/sdk/ad-targeting-ai.ts` ,

backend/functions/adTargetingLearn/ , backend/functions/adTargetingAiStatus/ ,
src/components/ads/AdMedia.jsx .

- **Changed:** backend/sdk/interstitial-ad.ts (targeting filter + media + AI ranking),
backend/functions/adGridFeed/entry.ts (PPC targeting + media + AI ranking),
backend/functions/createAdGridAd/entry.ts (store media + targeting),
backend/functions/socialAmplifyDistribute/entry.ts (cohort filter),
backend/functions/premiumPPCAutoAdvertise/entry.ts (cohort filter — gap fix),
backend/functions/premiumAdFree/entry.ts (pass user for targeting — gap fix), backend/sdk/settings.ts (flags),
backend/db/schema.sql (AdTargetingModel), backend/functions/_manifest.json , src/pages/AdGridSurvey.jsx
(the single canonical grid — full-screen single-tap takeover: 30s looping watch gate → questions-beneath → server-
side submit → product Buy-Now page → swipe-to-next), src/components/ads/InAppInterstitialAd.jsx ,
src/components/surveys/SurveyInterstitialAd.jsx , src/components/advertiser/AdSignupForm.jsx (advertiser
format + targeting UI).

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Ad-network metrics + demographic & audience targeting (2026-09 update)

Full network-standard metric set (measured, never guaranteed). Advertisers now get the complete mobile-ad-network metric set, computed from real platform activity: **eCPM, CPM, IPM, CPP**, and the windowed **ROAS curve (D1 / D3 / D7 / D14 / D28 / D90 / D365)**, plus the publisher-side yield metrics — **fill rate, ARPDAU and retention (D1 / D7 / D28)**. Every figure is measured and shown with its basis; the platform never guarantees an ROI. (Engine: `ad-metrics.ts`; on-demand read: `adMetricsReport`.)

Tracked over time and AI-optimized. A scheduled sweep (`adMetricsSweep`) tracks each metric as history (`OptimizationSignal` + `AgentLearningMemory` — no new tables) and runs an AI optimizer that biases delivery from measured results. Every automatic action is reversible and routed through the autonomy kernel (`ad_optimization`): it auto-applies on trust or queues a human review, and it **never raises spend and never touches billing or payouts** (those stay permanently gated).

Demographic + new-vs-existing targeting. On top of the interest/behavior cohorts, advertisers can target **demographics — age range, gender, country and region** — and **audience type (new vs existing users)**, matched against users' own non-identifying profile data (never an individual). Advertisers set an `optimize_for` objective — **ROAS, new-user acquisition, or existing-user retention** — that steers the AI delivery optimizer. (Engine: `ad-audience.ts`.)